

Introduction to Multi-Processor System on Chip (MPSoC)

Use concepts such as:

- Parallelism
- Pipelining
- Resource sharing
- Scheduling

Use VHDL & apply advanced techniques to design a complex digital system for an FPGA.

Use EDA tools to design application specific MPSoC and to satisfy resource and timing constraints in FPGA systems.

Assessments

Lab	20%
Research Project	20%
Group Research Project – ReCOP	30%
Group Research Project – Phase 2	30%

uP refers to microprocessor

MPSoC consists of interconnection network of uP, storage & custom IP to speed up processing

Examples of Processing Architectures for AI – from least to most efficient:
CPU – GPU – FPGA – Vector DSP – ASIC

Nios 2 (I think I forgot to save :/)

Profiling Nios 2 Systems

Performance Counters

High Res. Timers

GNU profiler

- performs periodic sampling to determine the percentage of time spent in each function.
 - an overview of the entire system
 - profiling overhead does not affect determining the most time consuming part of the code
 - uses system clock timer
 - binary code size increases
 - instruction cache behaviour may differ as instruction inserted for profiling
 - data cache behaviour different as results stored in memory
- pg compiler flag

Header files & Data types for C

Overview, Trends and Challenges of Modern Digital Design

"I/F": Interface

– trend towards high level synthesis

; tick separation

, separates RTL statements

Von Neumann Architecture: combined data instruction

Harvard Architecture: separated data and instruction memories

16 bit kdISA

– status bit

– 5 bits operator (32 instructions)

– 4 bit reg 1

– 4 bit reg 2

instructions

1. cmp
2. mov
3. swap
4. lli
5. lui
6. add
7. addi
8. sub
9. subi
10. or
11. ori
12. and
13. andi
14. not
15. xor
16. xori
17. test
18. load
19. store
20. srl
21. sll
22. sra

special registers

1. zero/null
2. program counter
3. instruction register
4. status register

32 bit kd ISA, want 32 registers

- 5 dst reg
 - 5 src reg
- 22 bits left
- 5 bit opcode

17 bits

status flags

- overflow
- zero
- negative
- carry

special registers

1. zero/null
2. program counter
3. instruction register
4. status register
5. stack pointer

MAR: Memory Access Register initialisation of registers

SystemC

- Pick and choose what *modules* are hardware or software.
- Modules can interface with each other
- SystemC simulations are synchronous
- Software modules are simulated independently (if on different cores)

SystemJ & SystemGALS

Disadvantage of SystemJ

- jvm

Group and Individual Project Goals

- Follows SystemJ concepts but separates on specification level control dominated from data dominated parts explicitly
- Uses *ReCOP* processors for control dominated parts in time critical applications
- General purpose processor and application specific processors are used for data dominated parts
- Network of Chip (NOC) uses TDMA-MIN
 - Conflict-less network (parts can communicate without interference)
 - Happens in one clock cycle